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Edge AI : From Smartphone to Smart Glass

1. What is Edge AI, and why now?
2. Smartphone Edge AI outlook
3. Smart glasses as the next Edge AI platform
4. Competitive implications and key takeaways

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Edge AI: Real-Time Intelligence at the Point of Action

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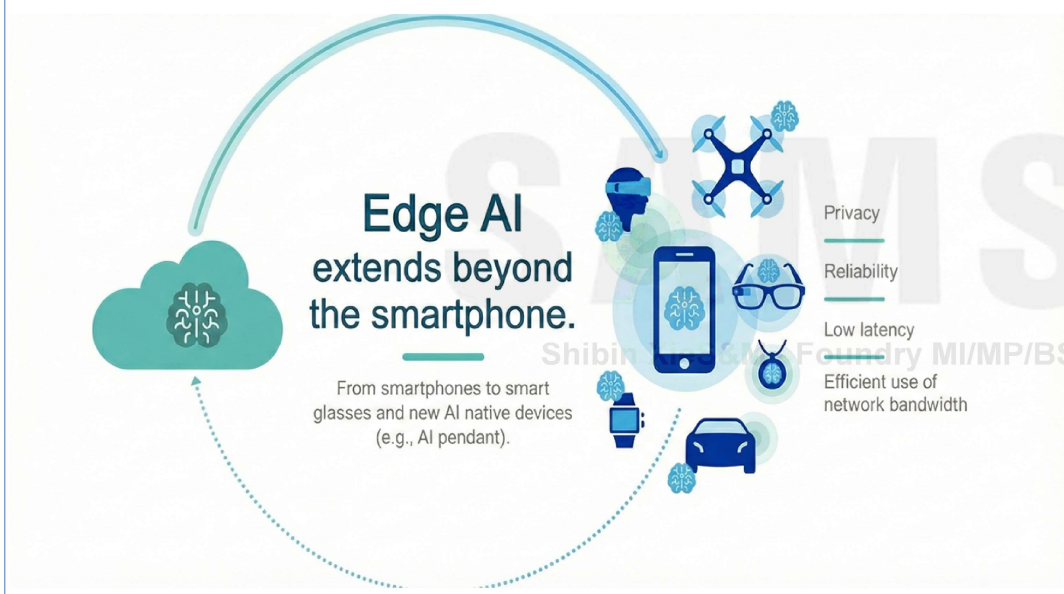
On-device AI handles immediate context-aware tasks while the cloud handles heavy compute workloads

Edge AI Execution Split: On-Device vs Cloud



- **Edge AI** runs inference **locally**, reducing latency, bandwidth use, and privacy risk
- **The cloud** remains essential for large-model processing, training, and orchestration
- Today, **smartphones** are the primary Edge AI platform at global scale
- The next phase shifts from **phone-first AI** to **AI-native** form factors

Source: TechInsights, DSA MI

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Edge AI: From Screen-Centric to Perception-Centric

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Edge AI is moving user interaction from app navigation to always-on, multimodal assistance

- Edge AI is shifting interaction from screen-centric smartphones to always-on, multimodal, context-aware companions
- This shift demands a silicon reset: lower-power AI-centric compute, advanced nodes, and tighter memory-logic integration under strict thermal limits

Form-Factor Evolution and Silicon Requirements

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Before 2024

GenAI Smartphone

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- Screen-centric
- 100–120mm² CPU/GPU/NPU die
- ~4–5W thermal envelope

2025 - 2026

Smart Glass

- Camera/audio-centric
- 40–60mm² CPU/GPU/DSP die
- Sub 1-2W always-on thermal envelope

2027 & beyond

Future AI Native Devices

LoveFrom,

- Multimodal, ambient AI
- Multi-die or chiplet-based
- 5–10W envelope with aggressive offload

Source: Jefferies, DSA MI

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